

# Sonjoy Kumar Paul

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## Summary

Third-year Ph.D. student at Texas A&M specializing in **Cybersecurity, Cognitive Security, and Brainwave Privacy**. Published at **USENIX Security** and **APWG eCrime**, with 5 additional papers under review. 5 years of professional experience as a C++ software engineer in large-scale financial software (2M+ LOC) used by Fortune 500 companies. Strong expertise in algorithms, applied machine learning, and systems security research.

## Research Interests

Computer & Network Security; Usable Security; End-to-end encryption (E2EE); Neuro-Security; Brainwave Privacy

## Education

- **Texas A&M University**, College Station, TX Jan 2024 – Present  
Ph.D., Computer Science (SPIES Research Lab, Advisor: Dr. Nitesh Saxena)
- **University of Michigan-Dearborn**, MI Jan 2023 – Dec 2023  
Ph.D., Computer and Information Science (Transfer Out)
- **Bangladesh University of Engineering and Technology (BUET)**, Dhaka Feb 2013 – Sep 2017  
B.Sc., Computer Science and Engineering

## Publications

- Mahbub Alam, Muhammad Lutfor Rahman, **Sonjoy Kumar Paul**, Amy W. Hays, Aftab Hussain, Md Imanul Huq, and Nitesh Saxena. “SoK: PHILTER: Uncovering Security and Functional Gaps in AI-based Phishing Website Detection Literature via an LLM-based Reasoning Framework,” in the 35th USENIX Security Symposium, August 2026.
- Morium Akter Munny, Mahbub Alam, **Sonjoy Kumar Paul**, Daniel Timko, Muhammad Lutfor Rahman, and Nitesh Saxena, “Infrastructure Patterns in Toll Scam Domains: A Comprehensive Analysis of Cybercriminal Registration and Hosting Strategies,” in 2025 APWG Symposium on Electronic Crime Research (eCrime), December 2025.

## Under Review

- **Sonjoy Kumar Paul**, Md Imanul Huq, Mahbub Alam, Amy W. Hays, Nitesh Saxena. “COGNI-WATCH: Did They Actually Read It? Detecting Security Warning Miscomprehension via Gaze and Cursor Dynamics”
- Md Imanul Huq, **Sonjoy Kumar Paul**, Aftab Hussain, Amy W. Hays, Mahbub Alam, Nitesh Saxena. “Tired and Phished: A Cognitive Security Study to Measure the Impact of Cognitive Fatigue on Phishing Website Detection Capability”
- Anuradha Mandal, Cagri Arisoy, **Sonjoy Kumar Paul**, and Nitesh Saxena. “*BrainPhone*: A Passive Side-Channel Attack for Speech and Identity Inference Using Consumer BCI Headsets”
- Md Shahidur Rahman, **Sonjoy Kumar Paul**, and Nitesh Saxena. “When Seeing Deceives: Compromising Multi-Device Linking in End-to-End Encrypted Messaging”
- Mahbub Alam, Aftab Hussain, **Sonjoy Kumar Paul**, Amy W. Hays, Md Imanul Huq, Nitesh Saxena. “SoK: FUZZCHECK.AI: On the Limitations of DNN Fuzzing to Discover Security Failures”

## Technical Skills

Machine Learning (ML), Deep Learning (DL), Large Language Models (LLMs); Programming Languages: C, C++, Java, Python, JavaScript; Databases: Relational (RDBMS), NoSQL (Apache Cassandra), Object-Oriented; Cloud & Big Data: Amazon Web Services (AWS), Apache Hadoop, Apache Spark; Frameworks & Tools: Spring Boot, Serverless Computing, Docker, Kubernetes; ML/AI Libraries: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas; Version Control & Systems: Git, Linux/Unix.

## Work Experience

**Graduate Assistant-Research**, Texas A&M University Jan 2024 – Present

- Led EEG-based privacy attack research (**BrainPhone**): demonstrated passive speech/digit inference and identity snooping via consumer BCI headsets (under review).
- Discovered **novel QR-/phone-based device-linking attacks** on E2EE messengers (WhatsApp, Signal, Telegram); proposed *Passkey-Anchored Linking* defense (under review).
- Developing **Cogni-Watch**, a cognitive-aware security framework detecting warning miscomprehension in real time via gaze and cursor dynamics (under review).
- Co-authored SoK studies on **AI-based phishing detection** (USENIX Security 2026) and **toll scam infrastructure** (eCrime 2025).

**Graduate Student Research Assistant**, Univ. of Michigan-Dearborn Jan 2023 – Dec 2023

- Analyzed GitHub commit histories of serverless apps, detecting inefficiency patterns.
- Improved GPU utilization for compute-heavy serverless workloads (DL & data processing).

**Senior Software Engineer**, CodeCrafters International Ltd. (Investortools, Inc.) Jul 2022 – Dec 2022

- Enhanced a multi-format **template engine** (Word/Excel/Email) with robust expression parsing.
- Designed and led development of an internal **Auction Management System** for training.

**Software Engineer**, CodeCrafters International Ltd. (Investortools, Inc.)

Jul 2019 – Jun 2022

- Improved **Expression Parser** performance (core engine of the platform).
- Built a **User Folder Migration** system enabling 200+ clients to migrate reports/graphs seamlessly.
- Contributed features across reporting, CLI automation, and proprietary OODBMS.

**Software Developer**, CodeCrafters International Ltd. (Investortools, Inc.)

Jan 2018 – Jun 2019

- Developed secure **report folder permission system** with granular access control.
- Delivered GUI modules using MFC with multi-process and background functionality.
- Diagnosed/resolved critical bugs, improving system stability.

## Selected Other Projects

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### TANGO: Low-Resource NLP Data Augmentation

- Performed large-scale data augmentation (**7.8M sentences**, 65 languages) for low-resource NLP and validated output quality through human evaluation studies.
- Designed a **prompt-based LLM evaluation framework** measuring semantic preservation, error severity, and lexical diversity of augmented text.

### Adversarial ML: Malware Detection and Evasion

- Built resource-constrained malware classifiers ( $\leq 1$  GB RAM,  $\leq 5$  s/query) and developed evasive malware binaries under both **black-box** and **white-box** threat models.

### Contrastive Vision–Language Pre-training (CLIP)

- Systematically benchmarked optimizer–loss combinations (Adam, AdamW, SGD  $\times$  CLIP, CyCLIP, SogCLR, VICReg, On-lineCLR) on CC3M and MSCOCO; identified **AdamW + CyCLIP** as the top-performing configuration.

### Software Security Exploitation & Defense

- Implemented end-to-end attack chains (buffer overflow, ROP, format-string) and corresponding defenses (bounds checking, CFI, software fault isolation) using fuzzing and symbolic execution.

## Professional Service

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- **External Reviewer**, The Web Conference (WWW), 2026

## Awards

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- One-Way Travel Support for Higher Education in the USA 2023 - Bangladesh Sweden Trust Fund
- **2<sup>nd</sup> Runner-up**, Software Project Show, Int'l Conf. on Networking Systems and Security (NSysS), 2016.